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Reshaping Fintech: Unveiling Recent Developments on Fintech Integration

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Abstract

The term FinTech integration refers to a process of amending financial services with the assistance of technology for sure. In other words, the concept translates as the application of technology to the well-established financial system to achieve superior efficiency, accessibility, and overall client experience. This research paper explores the phenomenon of FinTech integration by conducting a systematic mapping study, offering insights into the current trends and future prospects of research into this dynamic area. The examination had a coverage from numerous databases, namely Lens, Scopus, Xplore, and ACM to conduct a comprehensive review of all FinTech-related publications published in the period from 2010 to 2024. As the acquired data indicated a vast increase in the number of FinTech studies, specifically in the domain of Fintech integration, Fintech development, and Fintech adoption. In particular, a bibliometric analysis was performed to collect data on 1,013 research articles from the Lens database, selecting the most relevant documents. Furthermore, the method allowed for the mapping of the temporal scope of FinTech research, offering insights into the most common most types of journals, prolific authors and nations, as well as revealing the most frequent keywords and research areas. Likewise, revealing into the keyword co-occurrence network was created using the software program VOS viewer to showcase the relationships between the various themes of the collected literature. As a result, the research shed light on the rapid expansion of FinTech research in recent years, exposing the most prominent scholars,



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publications, and countries, as well as identifying upcoming trends in FinTech integration research. Ultimately, the significance of this study is not only in its provision of a coherent guide for those interested in learning more about the FinTech domain. It provides the research roadmap for further exploration within this multifaceted area of the integration of financial technology and its effect upon all participants across the financial field.

Keywords

Fintech, Financial technology, Fintech Integration, VOS viewer, Bibliometric

Introduction

The abbreviation of financial technology (FinTech) signifies a broadly defined concept in which the newer internet technologies evolve and coexist with the finance business [1]. Its definition can also be very heterogeneous, even within industries where it can refer to the whole financial service supply chain [2]. In other words, we can define FinTech in two primary ways either: (i) the development of technology products and services and provided to existing financial institutions; or (ii) the provision of innovative financial products and services using newer technology [2]. Furthermore, companies that craft innovative digital solutions specifically designed for the financial services sector can also be part of the FinTech [3]. Additionally, it is used to express entirely new business models which disrupt and reshape quite considerably financial services delivered and accessed in the market [4]. Some definitions consider FinTech to be an entire industry dedicated to exploiting the potential of technology to improve financial processes [5]. In the academia, FinTech is also known as a domain that bridges across interdisciplinary knowledge based on finance, technology and innovation management [6]. In this respect, the innovative nature of financial products and technologies as a result of FinTech is enabling it to act as a force for change [7], with a view to reducing the information asymmetry across the financial sector [8]. Crucially, FinTech also has the potential to play a key role in financial inclusion, enabling people who would otherwise be left out in the cold to join the mainstream financial system. The term FinTech is widely accepted both in business and academia and is applicable to a variety of situations in diverse fields [9]. But what is in common for all of these applications is that they fall in the internet-based and financial space that is called as the FinTech [10].

Yet the core factor connecting these applications is that all are Internet-connected and financial in nature; in other words, these are all examples of FinTech. It is precisely this plurality in the development of FinTech which has also meant its examination in numerous fields. They considered FinTech as disruptive, especially for conventional banks in the payments space. Also, an agreement on the description and conceptual framework of Fintech has not been reached among academics and practitioners. This has resulted in a multi-dimensional development of the term in various meanings [11]. At present, Fintech research is classified according to the business dimension. They categorized Fintech research into General Fintech,



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Payment, Clearing & Settlement, Risk Management & Investment, Market Aggregators, Crowdfunding, Peer-to-Peer (P2P) Lending, Cryptocurrency, and Blockchain. Wu [12] identified the following four types of Fintech companies based on company development stage and on values that derive from innovation: incumbent financial institutions, new entrants, new value addition and new efficiency. They also observed that, of articles under review, accruing to new entrants were the highest, while explaining the efficiency of the existing financial institutions was the least. Aysan and Nanaeva [6] proposed the notion of the digital finance cube in terms of business administration and function. The conceptualization of Fintech was thematically broken down into business functions (e.g., financing, investments, payment), technology and technology-related concepts (e.g., blockchain, social networks, near-field communication), and types of institutions (e.g., Fintech firms, traditional service providers). This analysis allows us to clarify our understanding of FinTech and to draw strategic lines to follow in the future. As Janik, Ryszko [13], the review of previous and current research is important for predicting the future direction of the FinTech evolution. Accordingly, with the relatively inadequate understanding of FinTech, the present study moves in by heavily examining published FinTech studies. This study will be the first to synthesize, comprehensively and coherently, the multiple threads of FinTech research, which exist and will evolve [14, 15]. Using a solid FinTech categorization structure, the current study will perform a systematic mapping study. This paper critically examines the status quo of FinTech research and its evolving trajectory and then identifies potential avenues for future research. To achieve this, the following research questions are tackled in this paper [16, 17]:

- RQ1: What is the state of the art of FinTech research?
- RQ2: How mature is FinTech research?
- RQ3: What variety of FinTech are presented within FinTech research?
- RQ4: What are the possible future directions of FinTech research?

The RQs provided herein are extensions of the primary motivation of this examination (i.e. assessing the present state and future prospects for FinTech research).

RQ1 and RQ2 are aimed at establishing a first common ground for the wider FinTech research area. By addressing these questions, the study seeks to systematize the level of maturity in the area of FinTech research [18, 19]. RQ3 goes a step further to offer a systematic review of current Fin-Tech typologies. Indeed, such analysis will serve to offer a classification for use when one undertakes further FinTech studies. Lastly, RQ4 uses the findings obtained from RQ1-RQ3 to investigate emerging trends in FinTech research [20, 21]. This systematic mapping review is unique compared to past studies due to its in-depth numerical results for a wider range of FinTech research. Although previous mapping studies could provide an intuitive



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comprehension of research trends, this study goes beyond description. It then discusses in detail to evaluate the maturity of research and recommends future research.

This study makes a number of key contributions to the literature:

- **Current State Reporting:** The paper gives a statistical overview of typical trends, key authors and top countries in global FinTech research.
- **Maturity Level Analysis:** By examining the overall theme of research, it tests the maturity level of current FinTech research.
- **Five-Dimensional FinTech:** The paper synthesizes different aspects of FinTech into five dimensions, thus offering a comprehensive perspective of the general nature of FinTech and shedding light on the overall trend, and hence makes a contribution to understanding the changing and emerging domain of study.
- **Future Direction Exploration:** The paper makes a detailed analysis to investigate potential future directions for FinTech research and the results quality by taking the effect of the selected articles into consideration.

Methodology

1.1 A Systematic Mapping Study

A SMR provides a map to a field of research. This kind of review synthesizes the literature on a selected topic and draws relationships between different fields concerned with that topic. The purpose is not to go in-depth into the contents of the studies, but rather to classify them by predefined keywords. This results in a coded database that can be employed for further detailed analysis. Unlike systematic literature reviews, systematic mapping reviews focus on other aspects. Systematic literature reviews seek to combine and evaluate all available research about a given research question, whereas systematic mapping reviews aim to map out the research domain and describe the nature of the research. There are several advantages to the approach of a systematic mapping review. They offer a broader view of the research field and can be used to decide if one or more rigorous reviews on particular studies in the map are warranted. A mapping review can enable researchers to decide whether the current literature relates to their research questions and if there are resources available to carry out a systematic review. This study uses the process described in the Social Care Institute for Excellence (SCIE) Systematic mapping guidance.

The criteria for inclusion table detail the criteria for selection of articles for study related to Fintech. It makes sure that only appropriate journal articles with titles or keywords, such as Fintech, Financial Technology, and the like, are included. Also, the papers should be



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presented in English and published between 2010 and 2024. This serves to direct research toward recent and relevant academic contributions to the Fintech phenomenon, in a uniform language.

Table 1: Inclusion criteria

IC1 – Key Terms	Key terms: “Fintech”, “Financial Technology”, “FinTech”, “Fin-tech” are included in the title and keywords section
IC2 – Document	Only published journal articles are included
IC3 – Language	Language of articles are all English
IC4 – Time Period	2010 to 2024

In contrast, the table of exclusion criteria describes which articles are to be excluded in order to ensure the quality and focus of the study. Articles which do not focus on Fintech or Fintech & integration are removed for irrelevant content. Additionally, articles of full-text access are only included, and the papers with duplicate data and with too much secondhand data are removed to ensure originality. Finally, the authors also remove any articles with plagiarized or copied content to provide academic integrity and trustworthiness in the results of the study.

Table 2: Exclusion criteria

EC1 – Irrelevance	Articles that do not focus on FinTech or Fintech Integration
EC2 – Availability	Full text must be available
EC3 – Repeated Data	Articles containing excessive secondhand data
EC4 – Plagiarism	Articles containing copied data and information

1.1 Database and search terms

To get an extensive and high-quality overview of the field of FinTech research, four major digital databases were selected: Association for Computing Machinery (ACM) Digital Library, Institute of Electrical and Electronics Engineers (IEEE) Xplore, Scopus, and Web of Science (WoS). ACM Digital Library and IEEE Xplore: Both offer more precise results for scholarly articles in the area of science, but are particularly strong in tech-related FinTech articles. ACM Digital Library - The ACM Digital Library is a research, discovery and networking platform containing: the Full-Text Collection of all ACM publications, including journals, conference proceedings, technical magazines, newsletters and books; a collection of curated and hosted



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full-text publications from select publishers; the ACM Guide to Computing Literature, a comprehensive bibliographic database focused exclusively on the field of computing. On IEEE Xplore, a large number of indexed conference proceedings are available, thus enabling earlier detection of new research directions.

1.2 Search string strategy

Systematic mappers should privilege coverage over depth. They strive for a wide coverage of a research area by surveying a large number of related papers. This methodological approach was adopted in this study, and the search strategy was based on titles and keywords. Main keywords used were “Fintech”, “Financial Technology”, “Fin Tech” and “Fin-tech”. This combination provided results that were slanted towards business-oriented FinTech research rather than within a technical or engineering orientation. Furthermore, to focus on a specific scope of literature, articles only dedicated to blockchain and cryptocurrencies were removed.

Table 1 (original design) probably represents how these key words were combined with the inclusion criteria to produce the search strings. Every database has its specific search syntax, and we had to modify our search strings for each platform independently. Table 2 (from the original article) probably gives you an idea of what the search string was like for the Scopus check.

Screening of research

The following article describes several steps involved in the screening. The first search process across the four selected databases produced 3064 FinTech articles. Duplicated articles (mostly Scopus and Web of Science) were eliminated from EndNote, leaving 1,013 articles. An additional 106 duplicates (pre-published articles) were further removed based on prespecified exclusion criteria in the manual screening process. This manual process helped to make the results useful and unambiguous later.

2.5 Preliminary Article Analysis

Table 3 shows the main bibliometric data of the review. It consists of all 1013 documents derived from 26 sources spanning from 2010 to 2024. A total of 17,906 keywords were extracted from the authors, and 2674 unique authors were included in the review. Of these, 88 were authors of single-authored papers, and 2586 of multi-authored papers. There are a total of 134 single-authored papers. The documents in the dataset received an average of 42.49 citations. The table highlights patterns of collaboration with an average authorship of 0.38 documents, 2.64 authors per document and 3.77 coauthors per document, a strong trend of cooperative research within the set of the reviewed literature.

Table 3: Main information table about the review:



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Documents	1,013
Sources	26
Author's keywords	17,906
Period	2010-2024
Average citations per documents	42.49
Authors	2,674
Authors of single-authored documents	88
Authors of multi-authored documents	2,586
Single-authored documents	134
Documents per author	0.38
Authors per document	2.64
Coauthors per document	3.77

In the field of scholarly publications, computer science reigns supreme with a staggering 168 articles. This is followed closely by business, management, and accounting with 126 publications. Economics, econometrics, and finance come in a close third at 108 articles. Social sciences hold the fourth position with 86 publications, while engineering rounds out the top five at 70 articles. Interestingly, the 'Others' category boasts a significant 276 publications, highlighting the diversity of research endeavors.

The pie chart represents distribution of articles from different research publication sectors. Computer Science is in the first position with 17% of all papers, followed by Business, Management and Accounting which consists of 12%. While the S (Social Sciences) constitutes 8%, and the E (Engineering) offers 7%. Table 5: Other main sections are Decision Sciences (5%), Mathematics (3%), of Environmental Science (3%), Energy (2%) and Physics and Astronomy (1%). With the 27% balance, meanwhile, it is 'Other' research references, rather than others per se, that are smaller scales of the scope of research area. This figure demonstrates the major contribution of Computer Science and business-related disciplines to this trend of research publications.



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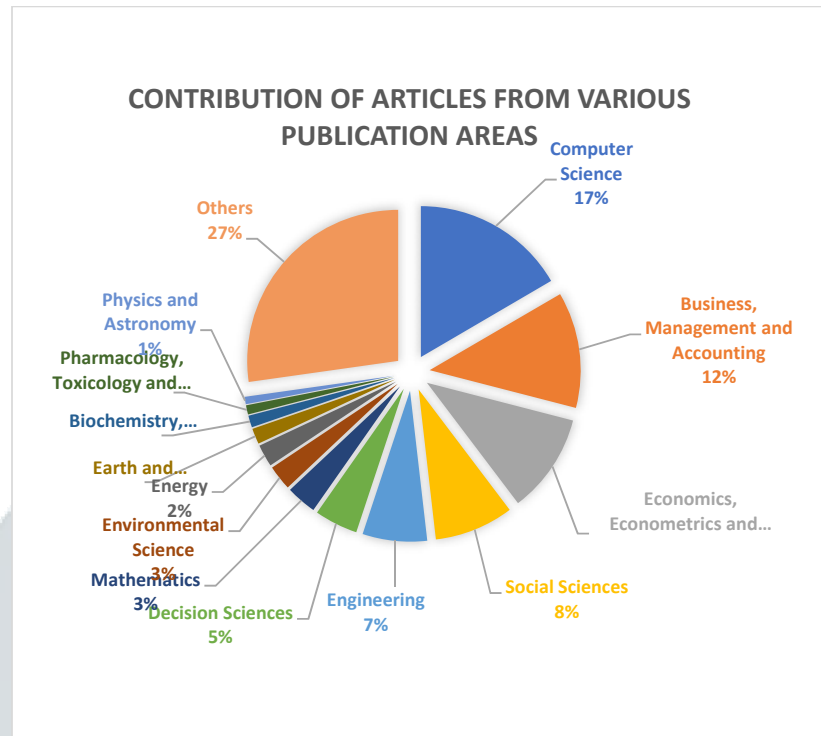


Figure 1: A pie-chart showing the number of articles from various research sectors

2.6 Keyword Cooccurrence Network

The map provides a meaningful visual of research topics and relationships in FinTech integration. There are a few key clusters and trends that emerge:

Financial inclusion: In this mainstream topic, “Financial inclusion,” “digital financial inclusion,” “financial performance,” and “financial development” are key topics. This presumably reflects a strong interest in how FinTech can improve access to financial services for unbanked people.

FinTech Adoption and Impacts The keywords ‘FinTech adoption’, ‘FinTech technology’, ‘FinTech landscape’, and ‘FinTech’ itself create another important cluster. This suggests an interest in understanding what is driving the take-up of FinTech, how it is influencing different sectors and what its overall impact on the financial ecosystem will be.

Technology and Innovation: The terms "technology", "blockchain", "artificial intelligence", "machine learning" and "innovation" are grouped, indicating the importance of technological development in shaping the FinTech industry.

Regulatory Challenges/Trends: The use of "financial regulation," "regulatory environment," "regulatory context," and "compliance" indicate that there are teams researching active regulators, how regulatory environments with FinTech investments evolve, as well as how to comply with the regulations.



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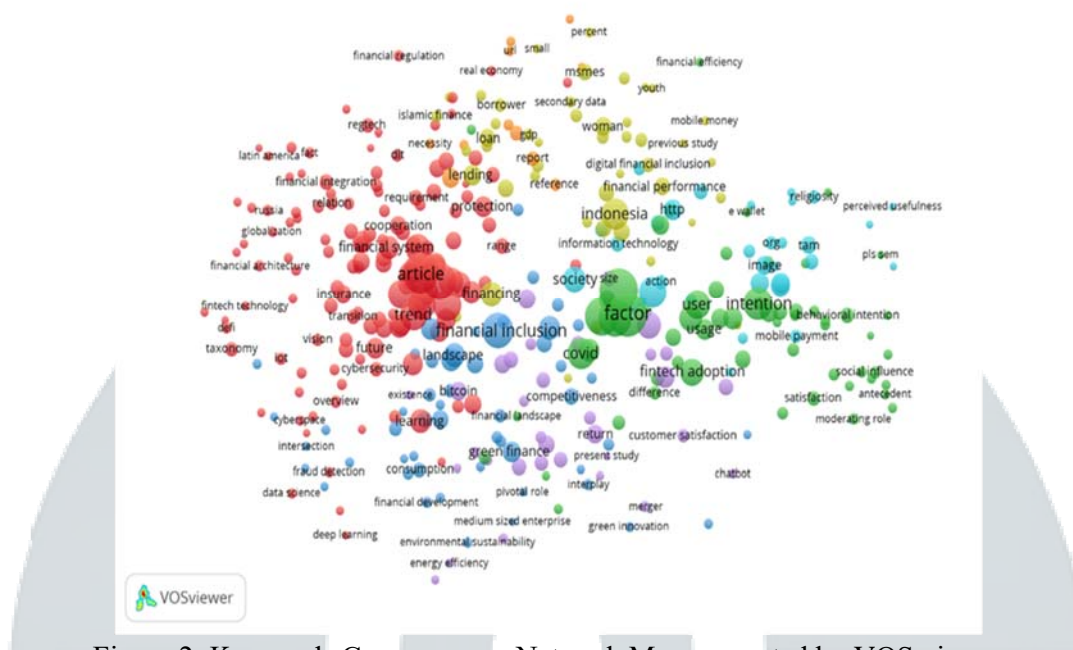


Figure 2: Keywords Cooccurrence Network Map generated by VOS viewer

Emerging Trends: There are also clusters around emerging trends such as “green finance”, “Islamic finance” and “cybersecurity”, which suggest some of those topics are attracting increasing research interest in the context of FinTech.

Limited Observations: And also, we observe that the map provides only the keywords mined automatically, from the analyzed papers. It gives no insight into the particular research questions, methods or results. Moreover, the frequency of keyword occurrence generally determines the number of clusters generated and their positions, which might not ideally reflect the relative importance of different research fronts.

The figure is an a word cloud plot which focuses on the academic fields and disciplines in which articles on "Fintech Integration" have been published, provided by the analysis of Lens. org. The size of each word or term represents the number or count of articles in that area of the journal foramen. Text size is positively associated with the volume of publications (Larger = more).

Dominant Field: Business (2,160 instances)

The most common category for fintech integration research is Business, suggesting that most of the literature related to fintech is of a business nature, such as market strategies, business models, and organizational effects.



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Finance (1,201) and Economics (886)

Drawing from the insights from these closely related fields, a great deal of interest is shown in financial services, economic implications, monetary policies, and market dynamics due to fintech-inspired innovations.

Computer Science (1,057)

The high proportion of computer science indicates tech -- particularly software development, algorithms, cybersecurity, blockchain, and AI in fintech -- is a big focus.

Other Prominent Fields:

Private sector financial services (782): Specific application spaces for fintech in banking, the payment system and related services.

Marketing (645): Emphasis on fintech product marketing, customer engagement and adoption strategies.

FinTech itself (541): Specific niche works in fintech; you could see both technology and fintech terms here.

Law (517): Inspired by legal background, policies, compliance, and governance for fintech.

Philosophy (342), Psychology (264), Biology (319):

Smaller but interesting presences that may signal the emergence of interdisciplinary works that relate fintech to terminal issues in those fields (e.g., behavioural finance, bioethics, cognitive dimensions of adoptions of fintechs, and possibly bioinformatics but with a fresh perspective on fintech).

Technical and Emerging Fields:

Artificial Intelligence (254), Machine Learning (217), Cryptocurrencies (130), Blockchain, Programming Languages, Software Engineering, Operating Systems, Structural Equation Modeling. These demonstrate the increasing technical depth of fintech research with advanced technologies.

Other social sciences and natural sciences:



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Disciplines like Sociology, Political Science, Geography, and even Chemistry and Ecology appear, which indicates the transversal character and the interdisciplinary nature of research in the field of fintech.

This figure shows that fintech integration research is multidisciplinary in nature, where researchers are found not only in business and finance but also in computer science, law, and with recent attention in AI, marketing, and social sciences. The number and size of these fields show where the academic focus is; they reflect how intricate and multi-faceted the fintech phenomenon is, as financial ecosystems and technology and society are blending.



Figure 3: Keywords and Number of Occurrences from Journal Articles on “Fintech Integration” by Lens.org Analysis

Results of Descriptive and Network Analysis

Scientific output, journals, authors, and geospatial analysis

Table 4: Top 20 Most Prolific Journals:

Journals	No. of Articles
Sustainability	107
SSRN Electronic Journal	98



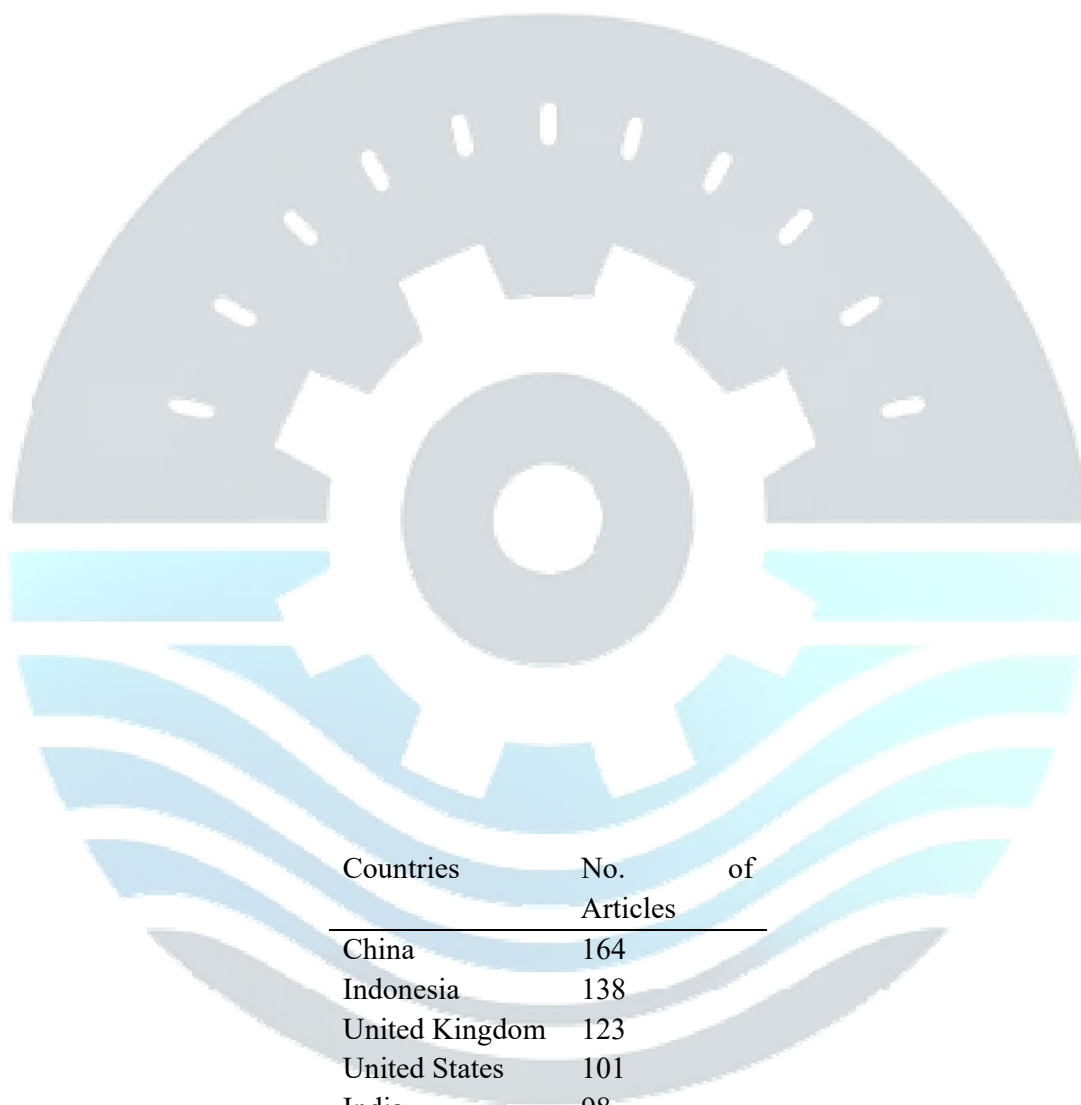
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FinTech	63
Financial Innovation	55
Journal of Risk and Financial Management	44
IEEE Access	25
Journal of Open Innovation: Technology, Market, and Complexity	25
IMF Working Papers	24
Financial innovation	21
Heliyon	21
Journal of Financial Services Marketing	19
Cogent Economics & Finance	17
Investment Management and Financial Innovations	17
E3S Web of Conferences	16
Electronic Markets	16
International Journal of Financial Studies	15
Risks	15
FinTech Notes	14
Frontiers in psychology	14
International Journal of Data and Network Science	14

Table 5: Top 10 Most Productive Countries



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Countries	No. of Articles
China	164
Indonesia	138
United Kingdom	123
United States	101
India	98
Germany	88
Italy	56
Saudi Arabia	44
Republic of Korea	29



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Taiwan	11
Total	852

Table 4 is a column chart that depicts the top 15 authors with the highest (filling in the value order) citation influence. It juxtaposes a pair of values: the count of papers written by each author, depicted as blue vertical bars and their link strength (shown as orange bars), which indicates the degree to which they are part of overall patterns of scholarly connectedness and influence in the academic citation graph. For instance, other writers, such as Ralf Weinad, have a lot of nodes and low link strength, meaning they generate a large number of documents but may not be as connected or central to cooperative networks. In contrast, authors like Xiaodie Pu and Rainer Alt present a very high link strength even though they have published less documents, which could reflect one of the two following scenarios, either the impact of their researches is very high with a massive quantity of citation belonging to collaborations, or their works appear as being influential and considered to have high reputation in their fields. Other authors like Gang Kou and Soumya Ray exhibit a more moderate profile with an approximately equal number of publications and link strength, which means they have shown reasonably consistent productivity as well as greater academic connectedness. This discrepancy between several documents and link strength illustrates different perspectives of academic influence: volume of production versus networked influence. The chart provides a general overview of how a scholar's visibility in his field of interest is influenced by either a large number of publications combined with potent collaboration or citation networks, and the strength of the links is an important factor reflecting scholarly impacts beyond the quantity of publications.



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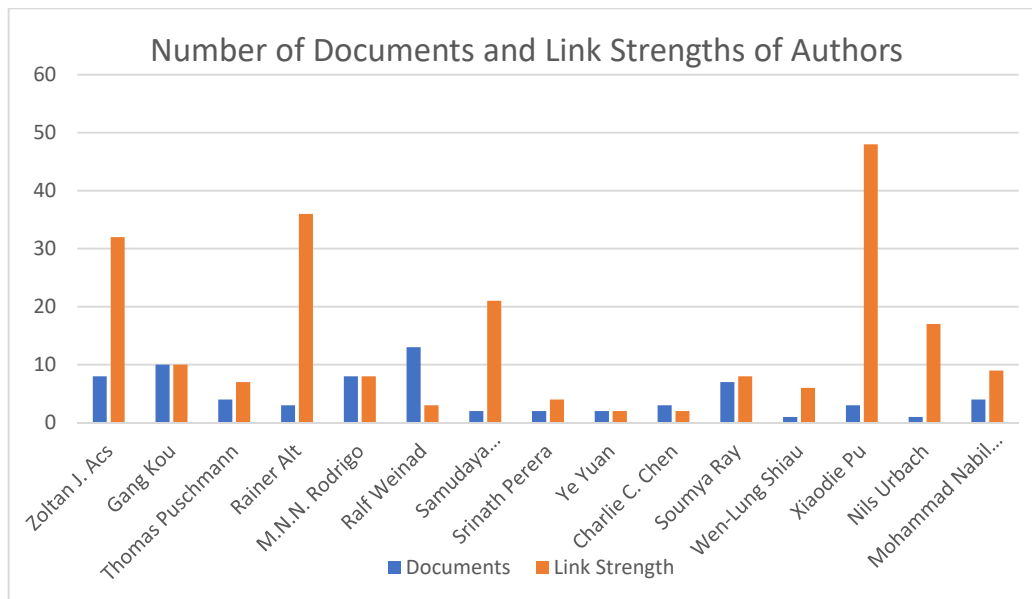


Table 4: A column-chart showing the top 15 Most Cited Authors (highest to lowest from left)

Figure 5: Published Journal Articles vs. Time Period

Figure 1 demonstrates the increase in journal articles published by year, from 2015 to the projected year 2025. It has progressively increased since 2018, but with a low publication volume from 2015 to 2017. The number of publications increases steeply, with more than 750 articles at the peak around 2022. This is an indication of increasing research interest and academic publications in this topic in recent years. The projected small drop in 2023 and 2024 may be because of incomplete data or a usual fluctuation after a peak, Dr. Ossi said. The increasing trend therefore likely indicates a growing level of academic interest in the topic, which may be as a result of new relevance, funding, or technological developments in this field.

Fig. 6: Article Citations vs. Date Published

This scatter plot illustrates how articles from various years have been cited over time. Each bubble is an article: its size being proportional to the number of citations it received. The articles that occupy a majority of the large bubbles are found to be mainly written between 2017 and 2019, indicating that they exercise an especially significant academic impact and influence. This is as would be expected, as older articles will have a number of citations per article as a function of age, unaltered from the original distribution. Moreover, some of the more recently published articles also have high numbers of citations, which indicates quick recognition and relevance to the field. Clustering of smaller bubbles around the more recent years may reflect a delayed accumulation of citations or a high number of articles that have been published more recently and not yet garnered many citations.



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Figure 7: Journal Article Published vs. Source

This graph classifies the total number of articles by journals, showing which journals have more contributions in the field. Several journals (chief among them, the journal Sustainability) drive significantly, with approximately 45 articles, soon followed by journals like IJERPH (International Journal of Environmental Research and Public Health) and Science of the Total Environment, with a large number of publications. Web sources like the Environmental Pollution Web-Alert may be added to the list as well. It is recommended that the list also include other well-reputed journals like Environment International, Journal of Cleaner Production and Environmental Science & Technology. The wide range of sources illustrates cross-sections with environmental science, public health and sustainable development. These are high-prestige journals that presumably act as important vehicles of dissemination for influential research in the field.

Fig. 8: Journal Article Published vs. Author

This graph shows the top-author distributions according to the number of journal papers they have published in this field. Wing-Kaung Wong is on #1, while Douglas A. Wrenn and Dariusz Wojciech perform well in the last two positions. It looks like these guys are big-wigs in the industry. Additional prominent authors who made the cut with high publication counts are also in the ranks, mirroring a division of research labor among a relatively small cadre. This information can provide insight on who the key experts are, whom to collaborate with or whose opinions are shaping research trends.

Ultimately, the figures together demonstrate a strong and increasing body of work in recent years with specific journals and authors dominating the scholarly output. Citation Patterns The pattern of what citations receive or give citations is of interest as it allows mapping the development of the field and the influence of the research. Such insights are useful for understanding the dynamic of publication, and for determining influential sources and leading researchers.



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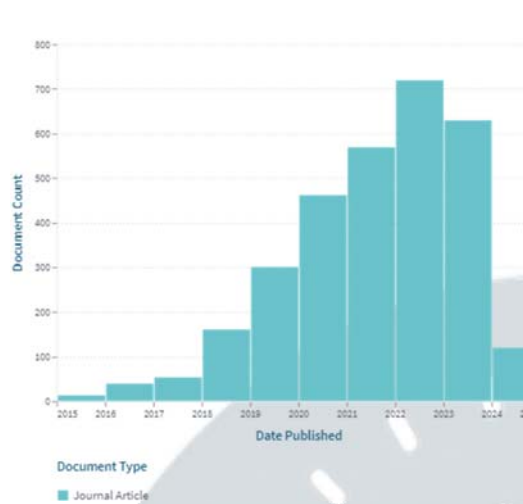


Figure 5: Journal Article Published vs. Time Period by Lens.org Analysis ⁱ

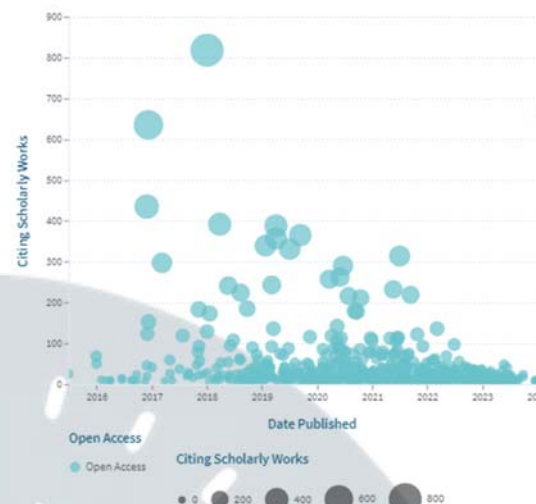


Figure 6: Citations of Articles vs. Date Published by Lens.org Analysis ⁱⁱ

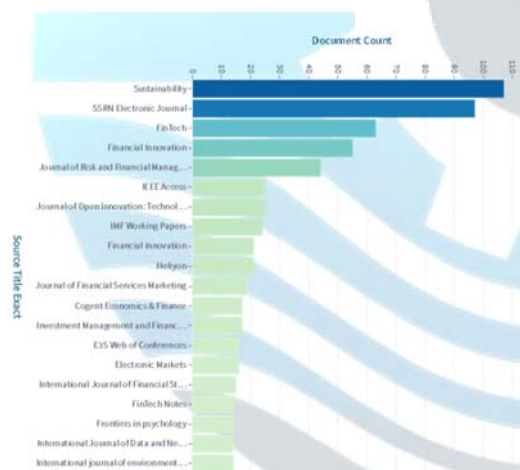


Figure 7: Journal Article Published vs. Source by Lens.org Analysis ⁱⁱⁱ

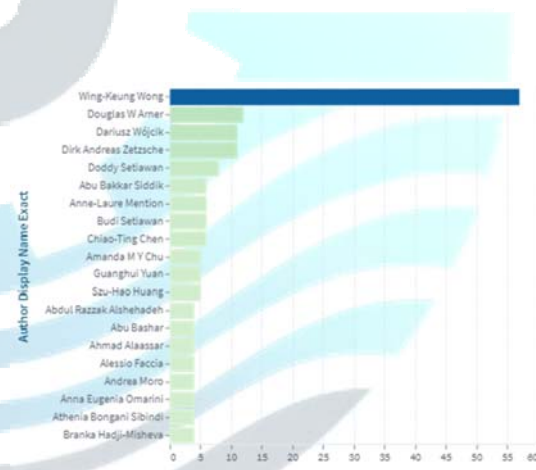


Figure 8: Journal Article Published vs. Author by Lens.org Analysis ^{iv}

Statistics of Keywords and Citations



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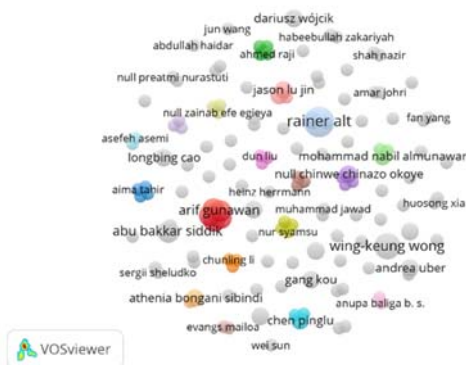


Figure 9: Co-authorship on Documents Cooccurrence Network Map generated by VOS viewer

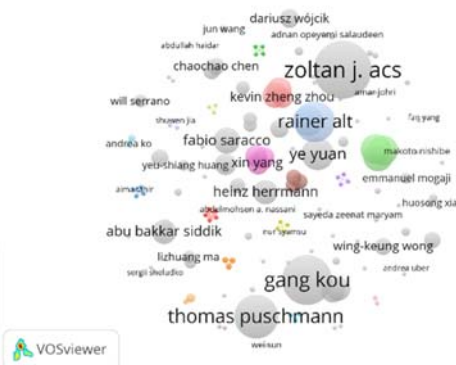


Figure 10: Co-authorship on Citations Cooccurrence Network Map generated by VOS viewer

These maps are graphical portraits of who collaborated with whom in a given dataset, which may be a collection of research papers (or publications). A circle (node) denotes an author and a line (edge) between circles indicates co authorship. The nodes are typically scaled to indicate the number of publications in the dataset that was authored. These maps show focused groups of authors indicating strong collaborative relationships. These are clusters of research communities, institutions, or fields of research. Certain authors, such as Rainer Alt and Jun Wang, are typically 'centrally positioned' in the network with many connections. It is probable that many of these authors have worked with various researchers, and this probably reflects the multi-faceted role they may have played or the active participation in several research projects.

Table 6: Keywords and Number of Occurrence from various Journal Articles



Keyword	Occurrence
FinTech	742
Development	604
Model	576
Integration	422
Finance	385
Innovation	363
Industry	358
Effect	346
Market	243
Economy	341
Challenge	320
Use	303
Application	301
Factor	298
Financial Service	284
Relationship	245
Sector	215
Energy	183
Process	158
Financial Inclusion	116

Discussion of Research Focus

The images display the results obtained from a bibliometric analysis carried out with the software tool VOSviewer, which is developed for the construction and visualization of bibliometric networks. Such networks are authors, citations of articles, authorship, co-occurrence of keywords, and other academic metadata. What are interesting to explore in this case are keyword clusters and link strengths of the keyword nodes which uncover the thematic structure of a research field.



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Figure 11: Keyword Clusters and Link Strength Network Map BO-L RAT Summary The results of the section 3.2. This map is a keyword network mapped from a research corpus. These keywords are grouped to one or more categories according to their association features, which is the frequency of co-occurring words. The node (keyword) size is proportional to the frequency or importance in the dataset. The keywords fall into two large clusters (red and green). The red cluster corresponds to a thematic community focusing on “development,” “finance,” and “inclusion”, which indicates a research focus on financial development and inclusion research questions. The green cluster represents theoretical and behavioral issues (as shown in keywords, such as “model,” “theory,” “intention,” “use”), which means studies on behavioral models and theories of the topic. Lines (connectors) between the nodes show the association strength between keywords more or thicker lines show higher connection/association strength or co-occurrence. The network has a high level of internal cohesiveness with clusters but also interconnected clusters indicating interdisciplinary/multi-dimensional research practice. The map represents splitting the research area in two channels. Development and Financial Inclusion: Focused on empirical and policy research on economic development and financial accessibility.

Behavioral Models and Theoretical Constructs - Based on theoretical contributions and studies in the field of technology acceptance and adoption. Figures 12 & 13: Closer-Up Cluster Maps. Figure 12: Cluster 01%Keyword: Development. This network map is an enlarged view of the map around the "Development" cluster in Figure 2 and the keywords that are centerpieces within this domain are emphasized. "The related keywords such as 'Finance', 'Inclusion', and 'Challenge' are emphasized, which demonstrates the main research foci here are solving challenges in finance inclusion and development areas.. Figure 13: Cluster 02 [Keyword: Model]. This map homes in on the “Model” cluster. It focuses on keywords on the theoretical frameworks and behavioural intentions, such as, “the-ory,” “use,” “intention and "behavior." This group emphasizes the analytical and conceptual aspect of the research, which deals with understanding and modeling user behavior and adoption, in a financial or an IT context.

The review reveals that the research landscape is twofold: practical usage (development, financial inclusion) and theoretical constructs (behavioral models, adoption theories). Despite identifiable clusters the fact they are interconnected evidences that research is also inter-field, in so far as theory is connected with practice. Researchers can use the prevalence and association of keywords to identify key themes and voids. Future research could investigate how (red cluster) financial inclusion strategies are better supported or explained by behavioral models (green cluster). These might point to or hint at new research areas or policy applications.



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Table 7: Term Cooccurrence and Keyword Clusters on Fintech Integration according to Keyword Cluster and Link Strength Map generated by VOSviewer

	Cluster 1	Cluster 2
No.	The role of Development on FinTech Integration	The role of different Models on Fintech Integration
1	Development	Model
2	Challenge	Use
3	Sustainability	Intention
4	Economy	Performance
5	Growth	Theory
6	Future	Society
7	Market	FinTech Adoption
8	Financial Inclusion	Trust
9	Feature	Application
10	Sector	Originality

1.1 Development's Role in Integrating FinTech is formed by their services.

1.1.1 Fintech Integration: A Revolution in Financial Landscapes for a Sustainable Future

The finance industry is in the midst of a revolution caused by the rapid growth and convergence of financial technology (FinTech). This fusion of people and cities presents challenges and opportunities, affecting everything from the economy to financial inclusion. This research paper explains FinTech and the state of FinTech integration, including the impact on different industries and on the creation of a sustainable financial world.

1.1.2 Growth and Evolution of the Market and Features

FinTech describes a myriad of technologies employed to provide financial services. There is certainly no shortage of innovative features, anywhere from mobile payments to online lending platforms to blockchain-powered investment solutions or AI-based wealth management offers



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when it comes to financial technology (FinTech). This feature innovation has enabled massive market growth, with the world's FinTech market set to become enormous in the future. Consumer demand for convenience, transparency, and access is fuelling this rise. FinTech provides services whose features meet these requirements, such as open access to finance (24/7 access to financial services), efficient transactions and personalized financial tools. Yet the breakneck progress also poses risks.

1.1.3 Challenges and Regulatory Environment

A critical obstacle in integrating fintech is the need for regulatory environments to evolve alongside new tech. Financial technology is changing so rapidly that it is hard for regulation to keep up. This introduces uncertainty to FinTechs and plain old FIs looking to adopt the tech. Moreover, concerns for privacy, consumer protection, and even financial stability are also factors. Regulators have to juggle encouraging innovation with having a sound financial system. Partnership with regulators is vital for a balance between innovation and managing its attendant risks, and between regulators and FinTech companies.

1.1.4 FinTech for inclusion and sustainability

Yet against these challenges, the promise of greater financial inclusion and sustainability remains huge. With technology, FinTech can tap into the underbanked audience to extend necessary financial services such as mobile banking and microloans. This has the potential to give individuals and communities an edge and stimulate economic growth and development. In addition, the Fintech solutions can support sustainability by enabling green investment and responsible finance behaviors. For example, it becomes possible to monitor and manage sustainable resources via blockchain technology. AI-driven tools can also scrutinise financial information to find and back the most environmentally friendly projects. The future of FinTech: Integrating Beyond Business By Robert EKESI Coordination Sector Programmed Manager, Trade, Investment and Innovation Programmed The future of FinTech: Integrating Beyond Business

1.1.5 FinTech's future: Sectoral impact and Sustainable Development

The FinTech penetration impact will be horizontal and far-reaching. In banking, FinTech is responsible for driving the wave of customer-centric and more efficient services to the fore. AI chatbots and automation combined can be a boon for simplified loan applications and customer support. Meanwhile, open banking APIs are encouraging cooperation between FIs and FinTech firms, giving consumers access to a greater array of financial products and services. InsureTech is also impacting large-scale disruptive trends in the insurance industry. Insur Tech start-ups have come up with products like parametric insurance, which collects real-time data to automatically be triggered to make a payment if very specific predictors are met. These solutions may be more customizable for insurance subscribers. With the eyes fixed on the



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horizon, the integration of FinTech could unlock a powerful source of development acceleration. FinTech products can also be used to drive a more informed consumer, promote financial education, educate investors (in a responsible way) and support sustainability. But maintaining this sustainable future depends on further cooperation among the parties involved.

1.2 Role of Various Models in FinTech Integration

1.2.1 Lifting the veil on FinTech Integration Fabric: A multi-layer approach

That FinTech integration, the embedding of financial technology into finance, is not as straightforward a matter as its popular conceptualization suggests, has the potential to have significant social implications. To understand why and how FinTech is adopted why is a theoretical discourse is required, and how is a methodological issue. We should present the development theories that direct it, the trust patterns that it creates, and the criteria for examining its performance. This research paper investigates the layering aspects of FinTech adoption, integration at different levels of a system architecture, from its origins, via intended use towards actual use.

1.2.2 The origin and foundation of object theory

FinTech was developed in the context of a widespread dissatisfaction with traditional financial services. Theories of economic efficiency and financial inclusion predict that outmoded systems typically do not serve the underbanked or provide services with the best possible levels of transparency and speed. FinTech was an answer to these issues, which used recent technological progress to make financial tools more user-centered, accessible and potentially cheaper. But the concept of FinTech isn't always just about efficiency. Such frameworks as The Diffusion of Innovation Theory suggest that acceptance of new technology is dependent on factors such as perceived ease of use, perceived relative advantage, and social influences. Understanding these drivers is important in predicting uptake rates of FinTech and in tailoring integration strategies to certain end-user segments.

1.2.3 Establishing Trust: A Critical Layer

Trust is one of the most important layers on the stack in FinTech [22, 23]. Financial services involve handling sensitive user information and personal financial security. And while FinTech solutions are convenient, users need to feel that their information is safe and that the application's products work as advertised [24, 25]. Trust must be built through transparency about how the data is handled and security is maintained. Regulatory considerations that ease these fears do play into building trust in FinTech solutions. And the source and brand of a FinTech company can exert a great impact on the trust level of its users as well [26, 27]. These types of organizations that have a heritage in established institutions may naturally have the lead here [28, 29].



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1.2.4 Applications and Performance Measurement

The potential of FinTech integration is huge and still growing. Types of FinTech There are countless types of FinTech, from mobile payment companies to online investing platforms to robo-advisors to blockchain technology [30]. To measure the quality of such applications, one has to use several metrics with care. The effectiveness of the integration of FinTech use, including user satisfaction, the quantity of transactions, and the rate of financial inclusion, is among the indicators of effectiveness. Furthermore, social impact figures such as the number of people with access to microcredit or taking up sustainable financial behavior can measure the social impact of FinTech solutions. But the novelty of FinTech applications cannot be disregarded. Some solutions may offer fresh new capabilities, whereas some others may offer a digital update or take on an age-old service. It would be garbage in (garbage out) if our approach weren't rooted in novelty, originality, for FinTech integration to yield real innovation and meet unmet segments of financial services.

1.2.5. The Social Fabric of FinTech

FinTech mixing is capable of redefining a societal tapestry. Financial inclusion. By improving access and reducing costs, FinTech can help to empower people and groups, drive economic growth and development, and promote social stability and sustainability. Innovatively, FinTech answers may also nudge financial literacy and well-balanced financial practices, resulting in a more financially stable society. There are, however, downsides as well. Given the potential for privacy violations and algorithmic bias in AI-driven applications, there is a growing need for ongoing oversight and responsible development guidelines. The societal consequences of FinTech adoption would depend on how successfully one is able to use it for good, as an enabler of good and a restrainer of bad.

Conclusion

This research underscores the ability of FinTech incorporation to be transformative in shaping the global financial scene as an innovative force facilitating financial inclusivity and sustainable economic development. Although FinTech is growing fast and has a wide range of use cases, there is still a lingering question about how to address issues such as regulation evolution, data safety and establishing user confidence. Our systematic map shows upcoming themes that focus on technology adoption, market development, and the societal implications of FinTech. Future studies can further explore the interaction between FinTech and the traditional financial system, analyze consumer behavior, and the optimal regulatory system that cross-marginalizes the conflict and the synergies that exist between innovation and protection. Moreover, investigating the links between FinTech and ESG opens up avenues towards reconciling technological progress with a broader socio-economic purpose. In the end,



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continued partnership between all stakeholders will be vital to unlocking the power of FinTech for a more inclusive and robust financial system.

One limitation of this work is that it only includes very fintech-centered articles. This implies that papers not related to FinTech, such as those that focus on blockchain or cryptocurrency, are excluded. Accordingly, more extensive research on these particular subthemes could be conducted for a more complete explanation. In that sense, FinTech incorporation is a disruptive agent in the finance industry. This integration is not without its obstacles, but there is great potential for reward. FinTech has the potential to play a role in stimulating economic growth, advancing financial inclusion, and driving a more sustainable future through innovation under an effective regulatory environment. With the FinTech landscape changing rapidly, constant institutional R&D is necessary to develop FinTech in a responsible and effective direction in all financial services applications. There are several key areas to be discussed in future research on FinTech. These are related to how FinTech can integrate easily with existing banking infrastructure, to what attitudes and behavior consumers have around FinTech service usage and to what regulatory framework is required to protect consumers in this industry. Moreover, the interaction between FinTech and sustainable development should also be investigated, given the significance of this phenomenon for the industry and society.

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